

Vector databases

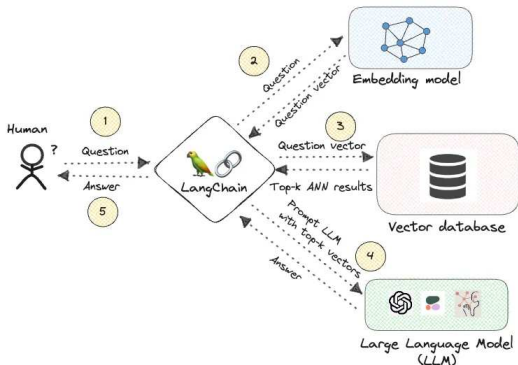
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September 22nd 2025, RH CZ AI Buddy, Brno

Retrieval Augmented Generation (RAG)

- Provides LLM accurate and relevant information for generating the response.
- Leverages external data source.
- Complementary strategy to LLM fine tuning how to improve relevance of the LLM's answers.



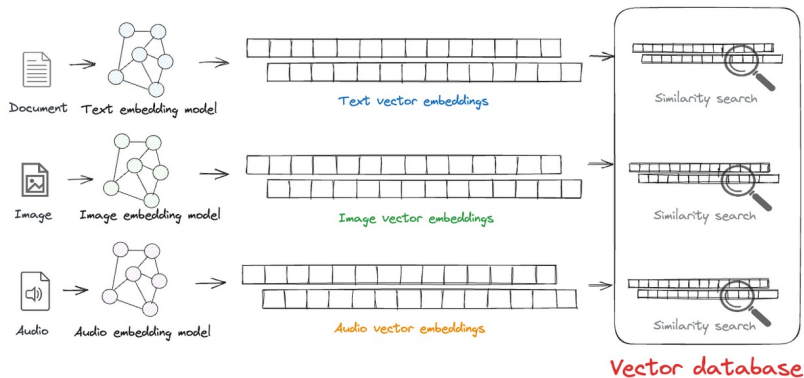
Source: <https://thedataquarry.com/blog/vector-db-2/>

Vector database

- Designed to store (high-dimensional) vectors.
- Effective search capabilities for similar vectors.
- Allow to store also vector metadata and/or original object from which the vector was derived.

Vector embedding

- Input data (text, audio, video) is transformed to high-dimensional vector by embedding model/transformer.
- Embedding captures some properties (usually semantics meaning) of the input data.
- Allows to do a similarity search across set of the vectors.



Vector embedding

- Similar embeddings correspond to objects which are also **semantically** similar.
- Vector embeddings have other interesting properties, e.g. well know

$$\textit{king} - \textit{man} + \textit{woman} \approx \textit{queen} \quad (1)$$

To be able to do similarity search, we have to define how to measure similarity
- we have to define metric on given vector space.

- Metric measure the distance of two vectors.
- It express similarity among the vectors.
- Usually smaller distance means bigger similarity, but there are also metrics where bigger number means bigger similarity.

Metric examples

- L2 (Euclidian)

$$d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

- L1 (Manhattan)

$$d(x, y) = \sum_{i=1}^n |x_i - y_i|$$

- scalar/inner product (vector need to be normalized)

$$x \cdot y = \sum_{i=1}^n x_i \cdot y_i$$

- cosine similarity

$$\cos \Theta = \frac{x \cdot y}{|x||y|}$$

- Jaccard index (similarity of two sample sets)

$$J(A, B) = \frac{|A \cap B|}{|A| + |B| - |A \cap B|}$$

- Enabled effective **similarity** search in high-dimensional vector space.
- As in relational DB, updating index adds overhead during data insertion.
- Some index types sacrifice precision for speed.

Vector indexes examples

- Inverted file (IVF)
 - Splits data into several clusters using e.g. K-means clustering.
 - Query identifies the most similar cluster and search only this cluster.
 - Has many variants like IVFFlat, IVFPQ (IVF product quantization), ...
- Locality-sensitive hashing (LSH)
 - Hash-based algorithm, vectors are clustered based on the fuzzy hashing function.
- Hierarchical Navigable Small World (HNSW)
 - Graph-based algorithm, nearest neighbours are connected.
 - Index organized in several layers, higher level are more sparse.

HNSW search

Vector indexes explained:

HNSW search

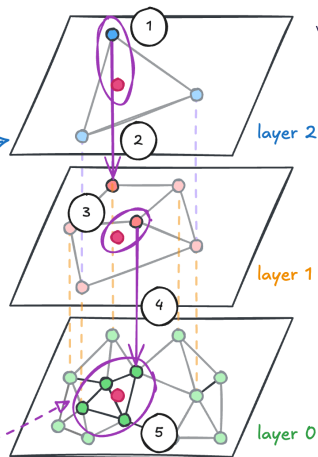
Vector search in HNSW proceeds by starting at the top layer, and proceeding down.

How would HNSW search for this vector: ● work?

Starting from the top layer,

- 1: find the most similar node,
- 2: travel down
- 3-4: repeat until bottom layer
- 5: then, find the closest k nodes

Search results



Source: <https://docs.weaviate.io/weaviate/concepts/vector-index>

Popular vector database

- All major relational databases now support vector data types and related functions.
- Major search engine supporting vectors



Elasticsearch



OpenSearch

- Specialized vector database



Chroma



Weavite



Pinecone



Milvus



Qdrant

- And many more ...

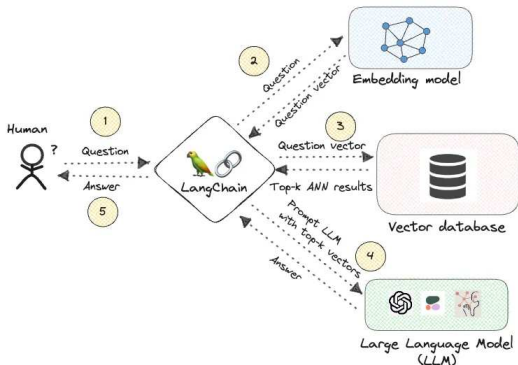
Vector Database Demo - pgvector

```
docker run --rm --name postgres -e POSTGRES_PASSWORD=postgres -d -it -p 5432:5432  
PGPASSWORD=postgres psql -h localhost -U postgres postgres
```

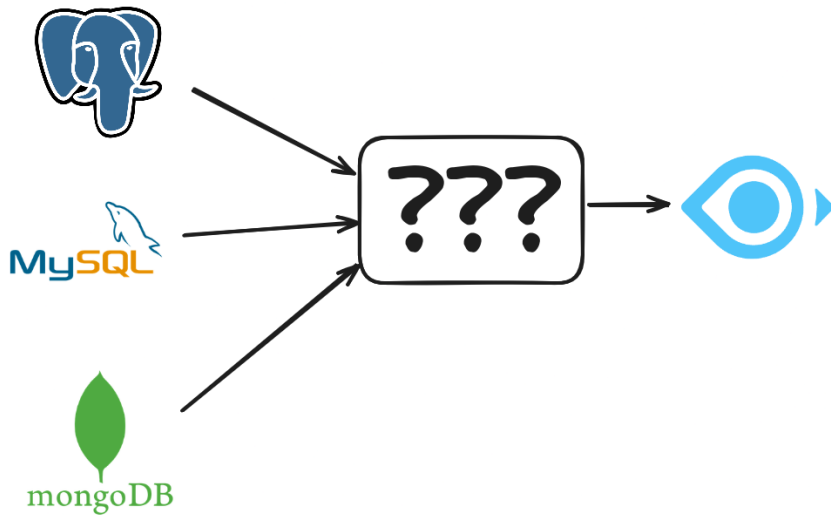
```
CREATE EXTENSION IF NOT EXISTS vector;  
CREATE TABLE vect_test (id serial PRIMARY KEY, embedding vector(3));  
INSERT INTO vect_test(embedding) VALUES ('[1,1,1]'), ('[2,2,2]'),  
('[3,3,3]'), ('[4,4,4]'), ('[5,5,5]'), ('[5,5,6]'), ('[5,6,5]');  
SELECT * FROM vect_test ORDER BY embedding <-> '[5,5,5]' limit 3;  
SELECT * FROM vect_test WHERE embedding <-> '[5,5,5]' < 2  
SELECT * FROM vect_test WHERE embedding <+> '[5,5,5]' < 2;  
SELECT * FROM vect_test WHERE embedding <=> '[5,5,5]' = 0;
```

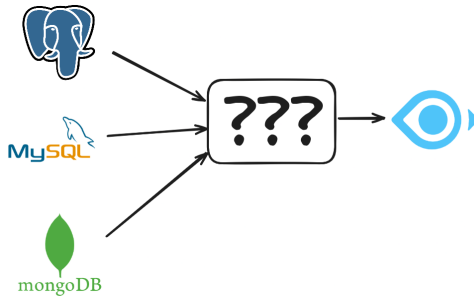
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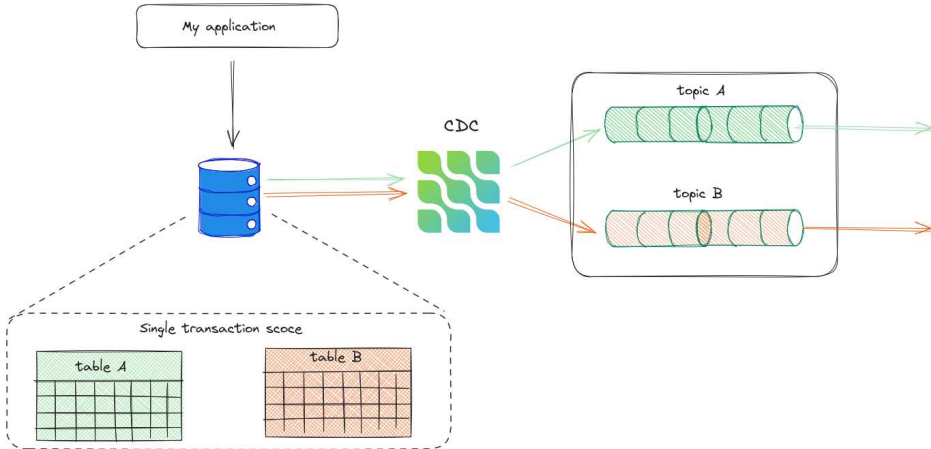
Source: <https://thedataquarry.com/blog/vector-db-2/>





- Consistent data, no data losses, no dual writes.
- Get all the changes without any delay in the real-time.
- Not overload the DB with the queries.

Change Data Capture (CDC)



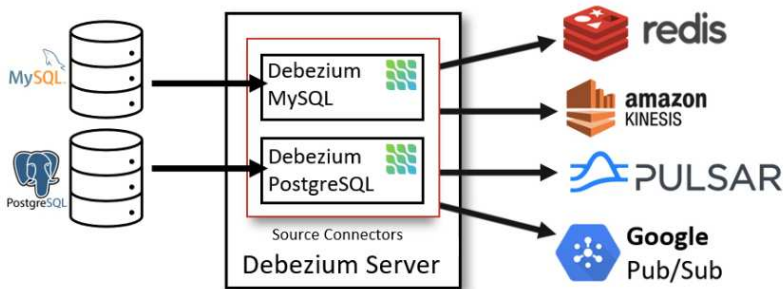


- Leading CDC framework, de-facto industry standard.
- Fully open source: <https://github.com/debezium/>
- Supports all major databases, including non-relational databases.
- Integrations with many 3rd-party tools and frameworks.
- Large and active user community.
- Used by many companies in production (see [Debezium public references](#)).

Debezium can be deployed as

- Kafka source connector
- standalone server
- library which can be embedded into Java application

Debezium as standalone server

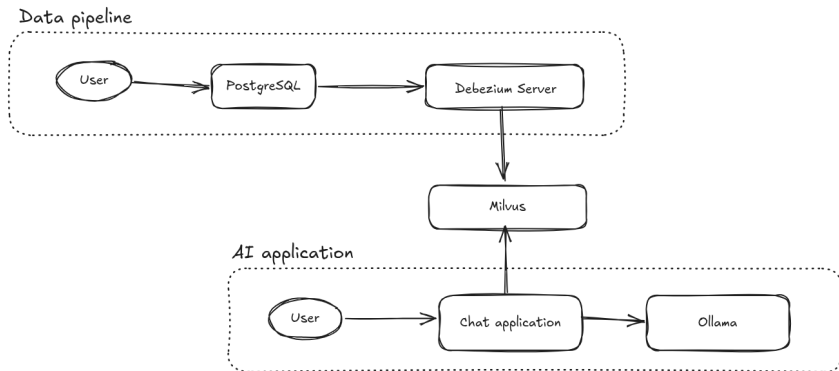


- Debezium provides AI module which allows to create embeddings as a single message transform (SMT).
- Integrated with major LLM providers.
- Currently Debezium server provides direct integration with Milvus, Qdrant and InstructLab.
- For Kafka Connect we provide JDBC sink connector, also dedicated Kafka Connect sink connectors for Milvus and others exist.
- No need to do any changes to the legacy applications or adjusting data flows.
- Modernization of the app stack can happen in parallel to existing applications without any distraction.

RAG Demo: PostgreSQL, Debezium, Milvus

See

- <https://github.com/debezium/debezium-examples/tree/main/ai-rag>
- <https://debezium.io/blog/2025/05/19/debezium-as-part-of-your-ai-solut>



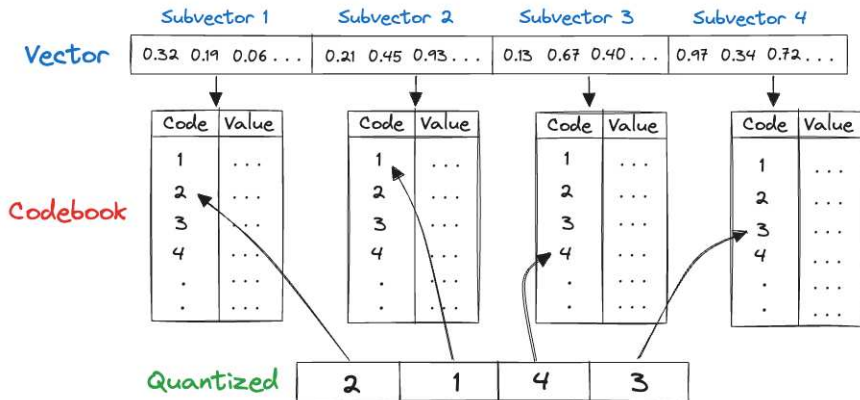
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Thank you!

Questions?

Backup slides

Product quantization



Source: https://lancedb.github.io/lancedb/concepts/index_ivfpq/